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Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 19	**Year: AY 2026-27**
TITLE : JDBC SELECT Operation	

Problem Statement

Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives

To establish database connectivity and retrieve records using JDBC SELECT statements.

Program:

sit_java.sql

```
CREATE DATABASE sit_java;
USE sit_java;

-- Table for Exercise 1
CREATE TABLE student (
   roll_no INT PRIMARY KEY,
   name VARCHAR(50),
   course VARCHAR(50),
   marks DOUBLE
);

INSERT INTO student VALUES
(101, 'Raghav Sarda', 'Computer Science', 92.5),
(102, 'Parth Kulkarni Sarda', 'Computer Science', 94.5),
(103, 'Yatharth Shah', 'Computer Science', 93.5);

-- Table for Exercise 2
CREATE TABLE product (
   product_id INT PRIMARY KEY,
   product_name VARCHAR(50),
   quantity INT,
   price DOUBLE
```

```
);

INSERT INTO product VALUES
(501, 'Logitech MX Master 4 Mouse', 10, 16000),
(502, 'Apple Magic Keyboard', 20, 18000),
(503, 'Apple Studio Display', 15, 179999);
```

StudentSelect.java

```
// Assignment 19 - Exercise 1: Display Student Records using SELECT
Query
import java.sql.*;
public class StudentSelect
{
   public static void main(String args[]) // Main method
   {
       String url = "jdbc:mysql://localhost:3306/sit_java"; // Database
URL
       String user = "thegharkalaptop"; // DB Username
       String pass = ""; // DB Password

       try // Exception handling for JDBC
       {
           Class.forName("com.mysql.cj.jdbc.Driver"); // Registering
JDBC driver
           Connection con = DriverManager.getConnection(url, user,
pass); // Establishing connection
           Statement stmt = con.createStatement(); // Creating Statement
object

           String query = "SELECT roll_no, name, course, marks FROM
student"; // SQL SELECT query
           ResultSet rs = stmt.executeQuery(query); // Executing query
and getting ResultSet

           System.out.println("== Student Records from Database ==");
           System.out.println("Roll No\tName\t\tCourse\t\tMarks");

           while(rs.next()) // Iterating through cursor row by row
           {
               int roll = rs.getInt("roll_no");
               String name = rs.getString("name");
               String course = rs.getString("course");
               double marks = rs.getDouble("marks");
```

```
                System.out.println(roll + "\t" + name + "\t" + course +
"\t" + marks); // Printing row
            }

            rs.close(); // Closing ResultSet
            stmt.close(); // Closing Statement
            con.close(); // Closing Connection
        }
        catch(Exception e) // Catching SQL and Driver exceptions
        {
            System.out.println("Database Error: " + e.getMessage());
        }
    }
}
```

ProductSelect.java

```
// Assignment 19 - Exercise 2: Connect to DB and display product details
using SELECT
import java.sql.*;
public class ProductSelect
{
    public static void main(String args[]) // Main Method
    {
        String url = "jdbc:mysql://localhost:3306/sit_java"; // Database
URL
        String user = "thegharkalaptop"; // DB Username
        String pass = ""; // DB Password

        try // Try block
        {
            Class.forName("com.mysql.cj.jdbc.Driver"); // Registering
driver
            Connection con = DriverManager.getConnection(url, user,
pass); // Connecting to DB
            Statement stmt = con.createStatement(); // Creating statement

            String query = "SELECT product_id, product_name, quantity,
price FROM product"; // SELECT query
            ResultSet rs = stmt.executeQuery(query);

            System.out.println("== Product Inventory List ==");
```

```
            System.out.println("ID\tProduct Name\t\tQuantity\tPrice
($)");

            while(rs.next()) // Traversing ResultSet
            {
                int pid = rs.getInt("product_id");
                String pname = rs.getString("product_name");
                int qty = rs.getInt("quantity");
                double price = rs.getDouble("price");

                System.out.println(pid + "\t" + pname + "\t" + qty +
"\t\t" + price); // Displaying record
            }

            rs.close(); // Closing ResultSet
            stmt.close(); // Closing Statement
            con.close(); // Closing Connection
        }
        catch(Exception e) // Catching exceptions
        {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output:

StudentSelect.java & ProductSelect.java

```
Problems  Output  Debug Console  Terminal  Ports

thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL % javac StudentSelect.java && java -cp ".:lib/*" StudentSelect
== Student Records from Database ==
Roll No Name            Course          Marks
101     Raghav Sarda    Computer Science        92.5
102     Parth Kulkarni Sarda    Computer Science        94.5
103     Yatharth Shah   Computer Science        93.5
thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL % javac ProductSelect.java && java -cp ".:lib/*" ProductSelect
== Product Inventory List ==
ID      Product Name            Quantity        Price ($)
501     Logitech MX Master 4 Mouse      10              16000.0
502     Apple Magic Keyboard    20              18000.0
503     Apple Studio Display    15              179999.0
thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL %
```

Conclusion:

In this assignment, database connectivity and sequential data retrieval using the JDBC SELECT operation were successfully implemented and verified. The DriverManager class established a live session with the MySQL database, the Statement interface executed SQL query strings, and the ResultSet interface enabled row-by-row cursor navigation to extract and format tabular data.